University of Warwick

Module Code: IB94X0 Business Statistics
School Year: 2026/27

Individual Assignment 1 Question and Guidance

The Structure

The assignment should be completed in RMarkdown and knitted into a html file. Your submission should be a zip file containing the RMarkdown file and the html output it creates. When the RMarkdown file is re-run, it must produce the same html output that you submit.

This should have two parts, as indicated in template_assignment1.rmd.

The first part should include the code that you produce to perform all stages of the data analysis. It should read in the datafile provided, check the integrity of data, calculate all of the statistics requested, and produce all of the figures/visualisations requested. The purpose of this first part is that it could be shared with someone else who is not familiar with the data or project, but is expert in R, tidyverse, and statistics. They should be able to see how you achieved your results, and if necessary, easily modify the code to work with new or updated data. Therefore, the appropriateness of your comments and data documentation will be assessed, as well as clarity and consistency of approach such as consistent use of compatible packages.

The second part should be a short, polished and professional report presenting and interpreting the findings for an audience of managers or policy makers. You should assume that the managers/policy makers have a basic knowledge of the data and where it comes from, but are not experts in statistics or R. Therefore, in addition to reporting the results of statistical tests using standard reporting conventions, you should also state the results in plain English and say what they mean or what they tell us about the real scenario – in this case what they tell us about the temperature on the different London Underground Lines. The visualisations and figures in this section should be as clear as possible and of publication standard.

The official word count for this assignment is 1500 words, but the written elements of the report **should be much shorter than this**. Since this is a code and analysis-based assignment, the word count is not directly applicable in the way that it is for an essay. The word count is provided as a guide: that this assignment is assumed to be approximately the same amount of work as a 1500-word essay. The comments (in code and between code blocks) in the first part should be short, brief, and sufficient for another data analyst to quickly determine what you are doing and why. The conclusions and writing in your second part should be no more than **300 words**. If you include more than this then you are likely including irrelevant information. Including irrelevant information or irrelevant code will have a negative impact upon your grade.

The details of what you should do are provided on the next page.

The Request

London Underground Temperatures

We are taking the role of a policy maker in London who is running a pilot scheme and is deciding where to install a new air-cooling system on the London Underground Lines. To do this we are going to use monthly average temperatures from the London Underground Lines:

Load in the data and check it for integrity/quality.

Provide a summary table that shows the following for each London Underground line where temperature levels were measured:
- The mean temperature
- The standard deviation of the temperature
- The lowest temperature recorded for that underground Line
- The highest temperature recorded for that underground Line

The data set includes older London Underground Lines, these Lines are unusual in that they are built closer to the surface than the other lines – these are grouped together and called "Sub-Surface Lines". You should exclude these from the subsequent analysis.

Plot the distribution of temperature across all years in the dataset. This should show the distributions separated by the underground line where temperature is measured (i.e. for each Line). The plot should also clearly show a visual representation of the mean value and standard deviation for each Line.

Perform a t-test to compare the temperature on the Bakerloo Line to the Central Line.

Perform an additional t-test to compare the temperature on the Bakerloo Line to the Central Line only in the summer months (i.e. May, June, July and August).

What would you advise the planner about where they should trail a new air-cooling system; the Bakerloo Line or the Central Line?

Marking Criteria

Comprehension
Demonstrate understanding of the data and the purpose of the analysis steps taken. Appropriately interpret both the task and the implications of the results.

Analysis & Critical Evaluation
Apply appropriate checks and validation of the data and apply appropriate analyses. Where appropriate, critique and evaluate the data and results.

Academic Writing
Clear description of the analysis and results that is appropriate for the stated audience(s). Concise description that clearly identifies and communicates the important and relevant elements. High quality visualisations that demonstrate the desired information or results.

Technical Capability
Clear and appropriate data analysis with well structured code and appropriate commenting.

Maximum Length:	**1500 words** *(see main text above)*
Assessment Weighting:	**20%**
Submission Deadline:	**29 October 2026 12:00 (Midday, UK time)**
Marks Released by:	**Thursday 26 November 2026** *(we will aim to release marks by this date, but in the event of an unavoidable delay a message will be posted on the module page)*
Artificial Intelligence:	*PERMITTED*

PERMITTED (neither prohibited or required)

If you use a generative Artificial Intelligence (AI) in the process of completing this assessment you MUST set out clearly the following:

- WHY you used a generative AI
- WHAT it was used for
- WHICH AI was used; and
- If any generated content has been used directly in this submission, if so where.

Note that this declaration does NOT contribute towards the word count for the assessment.

You will also have to confirm in your declaration that the work remains yours and you have intellectual ownership of it. You may be called for viva or other interview to demonstrate such intellectual ownership. A failure to disclose the use of AI, or the use of a misleading description of its use may have significant consequences for your studies. As a result, keeping good records of your interactions is strongly advised.

For further guidance, please see you Student Handbook and the Assessment Success Tool in my.wbs

Submitting your work

Before you submit your assessment, you should ensure you are familiar with the guidance and rules in the "Your Assessments" section of your Student Handbook, paying particular attention to:

- Academic Integrity (including Plagiarism)
- Referencing
- Word Count Policy and Formatting

- Confidentiality of your work
- Extensions and late submission of work
- Guidelines for Online Submission

Mitigating Circumstances

Mitigating circumstances MUST be submitted **within 20 working days following the submission deadline**. Mitigating circumstances not submitted by the relevant deadline are not required to be considered by the School/Department and may have to be considered by an Academic Appeals Committee as part of an academic appeal – for further information, please see: https://warwick.ac.uk/services/gov/calendar/section2/regulations/reg42academicappeals